Department of Electrical Engineering
Indian Institute of Technology Kharagpur

Machine Learning for Signal Processing Laboratory
(EE69210)
Spring, 2023-24

Experiment 10:
Gender recognition using speech data

Grading Rubric

	Tick the best applicable per row			Points
	Below Expectation	Lacking in Some	Meets all Expectation	
Completeness of the report				
Organization of the report (5 pts)				
Quality of figures (5 pts)				
Creation of Dataset and explanation of the preprocessing steps (35 pts)				
Model training, inference, and explanation for the base model with discussions. (25 pts)				
Generation of the confusion matrix and discussion. (25)				
Discussion on how to improve accuracy. (5 pts)				
			TOTAL (100 pts)	

1 Code of Conduct

Students are expected to behave ethically both in and out of the lab. Unethical behaviour includes, but is not limited to, the following:

- Possession of another person's laboratory solutions from the current or previous years.
- Reference to, or use of another person's laboratory solutions from the current or previous years.
- Submission of work that is not done by your laboratory group.
- Allowing another person to copy your laboratory solutions or work.
- Cheating on quizzes.

The rules of laboratory ethics are designed to facilitate these goals. We emphasize that laboratory TAs are available to help the student understand the basic concepts and answer the questions asked in the laboratory exercises. By performing the laboratories independently, students will likely learn more and improve their performance in the course as a whole. Please note that it is the student's responsibility to ensure that the content of their graded laboratories is not distributed to other students. If there is any question as to whether a given action might be considered unethical, please see the professor or the TA before you engage in such actions.

2 Gender Detection using speech.

Speech recognition technology is pivotal in gender detection by analyzing vocal patterns and characteristics. As individuals speak, their voices exhibit unique features influenced by biological factors such as vocal cord length and pitch. Advanced algorithms embedded in speech recognition systems can identify these distinctive traits to infer the speaker's gender accurately. By leveraging machine learning techniques, these systems continuously refine their models based on diverse voice samples, allowing for improved accuracy and adaptability across various languages and dialects. This technology has broad applications, from enhancing virtual assistants' understanding of user preferences to aiding law enforcement in voice profiling for investigative purposes.

Practically, gender detection through speech recognition is seamlessly integrated into numerous applications. Virtual assistants like Siri, Google Assistant, and Alexa utilize this technology to tailor responses and interactions according to the perceived gender of the user. Additionally, in sectors such as customer service and market research, speech-based gender identification assists in customizing communication strategies. However, ethical considerations related to privacy and potential biases in algorithmic models underscore the need for ongoing research and development to ensure fair and responsible implementation of gender detection through speech recognition technology.

2.1 The need for gender detection.

Gender detection serves various purposes across different domains, contributing to various applications that enhance user experiences, optimize services, and address specific needs. Here are some reasons why gender detection is considered important in various tasks:

1. **Personalization of Services:** Gender detection is crucial for personalizing services and interactions in applications such as virtual assistants, chatbots, and customer support systems. Understanding the user's gender allows these systems to tailor responses, recommendations, and content to suit individual preferences and needs better.

2. **Targeted Marketing and Advertising:** In the realm of marketing and advertising, knowing the gender of users helps in creating more targeted and relevant campaigns. Advertisers can tailor their messages, products, and promotions based on the demographics of their target audience, leading to more effective and efficient marketing strategies.

3. **Healthcare and Fitness Applications:** In healthcare and fitness applications, gender detection can be essential for providing personalized health advice, fitness routines, and nutritional recommendations. Men and women may have different health needs, and recognizing gender allows these applications to offer more accurate and customized guidance.

4. **Security and Authentication:** Gender detection can play a role in security applications, especially in voice-based authentication systems. Adding gender as a factor can enhance the accuracy of biometric authentication, making it more challenging for unauthorized individuals to gain access to sensitive information or facilities.

5. **Social and Educational Platforms:** Social and educational platforms may use gender detection to create inclusive and supportive environments. This information can help tailor educational content, promote diversity, and prevent the spread of gender-based harassment or discrimination.

6. **Demographic Analysis and Research:** Sociological and demographic research often relies on gender data to understand and analyze different populations' trends, behaviours, and patterns. This information is valuable for making informed decisions in public policy, healthcare planning, and social development.

While gender detection can benefit these areas, it is important to approach this task with ethical considerations, ensuring that privacy is respected and potential biases in the algorithms are addressed. Striking a balance between the advantages of gender detection and the responsible use of this information is crucial for its continued importance and acceptance across various applications.

2.2 Speech Features

- **Mel Spectrogram:**

  $$\text{Mel Spectrogram} = \text{TriangularFilterbank} \times \text{STFT}$$

  The Mel spectrogram represents the spectrum of a signal, where the frequency axis is converted to the Mel scale. It is obtained by dividing the audio signal into short time frames using the Short-Time Fourier Transform (STFT) and transforming each frame's spectrum into the Mel scale using a filter bank of triangular filters.

- **MFCC (Mel-Frequency Cepstral Coefficients):**

  $$\text{MFCC} = \text{DCT}\left(\log\left(\text{Mel Spectrogram}\right)\right)$$

  MFCC is a feature extraction technique commonly used in speech and audio signal processing. It involves computing the Mel spectrogram, taking the logarithm of the spectrogram to approximate the human ear's response, and applying the Discrete Cosine Transform (DCT) to obtain a set of coefficients. MFCCs effectively capture essential features of audio signals and are widely used in tasks such as speech recognition, speaker identification, and audio classification.

2.3 Dataset Description

The dataset used for this experiment is Mozilla's Common Voice Dataset, a corpus of speech data read by users on the Common Voice website. Its purpose is to enable the training and testing of automatic speech recognition, but with the information provided in the dataset, we can use it for gender recognition. All the audio files in this dataset are sampled at 48kHz and the average length of the utterance is around 4 seconds.

2.4 Architecture of the Fully Connected network

The following network is to be created and deployed. It has five layers including one output layer.

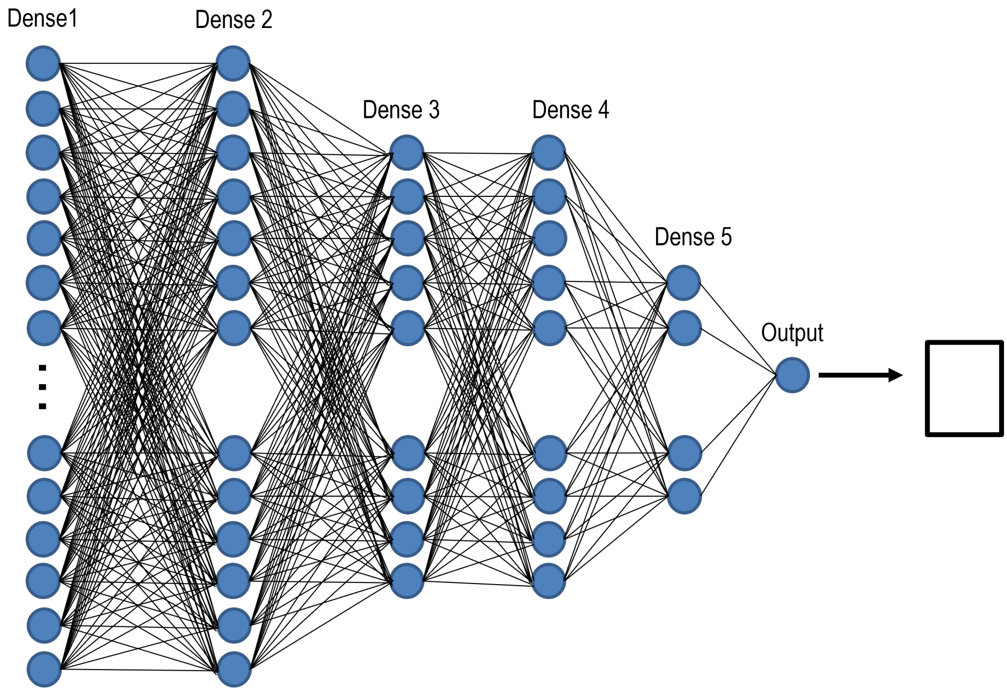

Figure 1: The neural network for this lab manual features five fully connected layers, denoted as Denser. The final output layer consists of 10 neurons, each corresponding to a specific class of the dataset, providing a structured representation for classification.

1. **Dense 1 :**
   - **Fully Connected :** This is the first layer of the network. being a fully connected layer, it takes in the input and has 256 output neurons.
   - **Dropout:** Probability of 0.3
2. **Dense 2 :**
   - **Fully Connected :** Fully connected layer with 256 output neurons.
   - **Non-Linearity :** The layer is followed by ReLU activation.
   - **Dropout:** Probability of 0.3
3. **Dense 3 :**
   - **Fully Connected :** Fully connected layer 128 output neurons.

- **Non-Linearity :** The layer is followed by ReLU activation.
- **Dropout:** Probability of 0.3

4. **Dense 4 :**
   - **Fully Connected :** Fully connected layer with 128 output neurons.
   - **Non-Linearity :** The layer is followed by ReLU activation.
   - **Dropout:** Probability of 0.3

5. **Dense 5 :**
   - **Fully Connected :** Fully connected layer with 64 output neurons.
   - **Non-Linearity :** The layer is followed by ReLU activation.
   - **Dropout:** Probability of 0.3

6. **Output :**
   - **Fully Connected :** This is the final and output layer of the network. It has one output neuron for each class of the dataset.
   - **Non-Linearity :** Sigmoid Activation is used.

2.5 Loss function

Cross-entropy loss, commonly used in machine learning for classification tasks, quantifies the difference between predicted probabilities and actual class labels. Specifically, in the context of a neural network, the cross-entropy loss measures the information gained with the model's predictions. The formula takes the negative logarithm of the predicted probability assigned to the correct class. A lower cross-entropy loss indicates that the model's predictions align well with the actual labels, while a higher loss suggests a greater divergence. Minimizing cross-entropy is a common objective during training as it encourages the neural network to converge towards accurate predictions. Mathematically, cross-entropy loss is given by

$$L(y, \hat{y}) = -\sum_{i=1}^{C} y_i \cdot \log(\hat{y}_i) \tag{1}$$

where y is the ground truth label vector and $\widehat{y}$ represents the predicted probability vector. As we have only two classes, $\hat{y}_1 = 1 - \hat{y}_0$. So if we replace $\hat{y}_1 = \hat{y}$ the equation thus becomes,

$$L(y, \hat{y}) = -y_0 \cdot \log(\hat{y}) - y_1 \cdot \log(1 - \hat{y}) \tag{2}$$

2.6 Learning Rate

The learning rate is a crucial hyperparameter determining the step size at which the model adjusts its weights during training. We will put the learning rate as 0.001, implying a moderate step size, allowing the model to adjust its parameters gradually. This balance is essential for avoiding overshooting or convergence issues. Experimenting with different learning rates can impact the training dynamics, affecting the model's ability to converge to an optimal solution.

2.7 Dropouts

1. **Dropout :**During training, the output of each neuron is set to zero with a probability p. This is mathematically represented as:

   $$Output = Input \times Bernoulli(p) \tag{3}$$

   where $Bernoulli(p)$ is a random binary mask with a probability p.

3 Experiments on Gender Detection using speech

3.1 Importing essential libraries

```
#importing libraries
import glob
import os
import pandas as pd
import numpy as np
import librosa
from tqdm import tqdm
```

Listing 1: Code to import essential libraries

3.2 Function to extract features from audio files

```
def extract_feature(file_name, **kwargs):
    """
    Extract mfcc and mel-spectrogram from audio file `file_name`
        e.g:
        `features = extract_feature(path, mel=True, mfcc=True)`
    """
    mfcc = kwargs.get("mfcc")
    mel = kwargs.get("mel")
    X, sample_rate = librosa.core.load(file_name)

    result = np.array([])
    if mfcc:
        mfccs = np.mean(librosa.feature.mfcc(y=X, sr=sample_rate, n_mfcc=40).T,
     axis=0)
        result = np.hstack((result, mfccs))
    if mel:
        mel = np.mean(librosa.feature.melspectrogram(y=X, sr=sample_rate).T,
    axis=0)
        result = np.hstack((result, mel))
    return result
```

3.2.1 Download and preparation of the dataset

First download the dataset from https://www.kaggle.com/datasets/mozillaorg/common-voice and then from the .zip file extract the folders and the .csv files named ”cv-valid-train”, ”cv-valid-test” and ”cv-valid-dev”. After the extraction of these files run the preprocessing code given below.

```

dirname = "newdata"

if not os.path.isdir(dirname):
    os.mkdir(dirname)

csv_files = glob.glob("*.csv")

for j, csv_file in enumerate(csv_files):
    print("[+] Preprocessing", csv_file)
    df = pd.read_csv(csv_file)
    # only take filename and gender columns
    new_df = df[["filename", "gender"]]
    print("Previously:", len(new_df), "rows")
    # take only male & female genders (i.e droping NaNs & 'other' gender)
    new_df = new_df[np.logical_or(new_df['gender'] == 'female', new_df['gender'
    ] == 'male')]
```

```python

    # the 0 is the label for the male and 1 is the label for the female voice
    new_df.loc[df["gender"] == "male", "gender"] = 0
    new_df.loc[df["gender"] == "female", "gender"] = 1

    print("Now:", len(new_df), "rows")
    new_csv_file = os.path.join(dirname, csv_file)
    # save new preprocessed CSV
    new_df.to_csv(new_csv_file, index=False)
    # get the folder name
    folder_name, _ = csv_file.split(".")
    audio_files = glob.glob(f"{folder_name}/{folder_name}/*")
    all_audio_filenames = set(new_df["filename"])
    for i, audio_file in tqdm(list(enumerate(audio_files)), f"Extracting
   features of {folder_name}"):
        splited = os.path.split(audio_file)

        audio_filename = f"{os.path.split(splited[0])[-1]}/{splited[-1]}"
        if audio_filename in all_audio_filenames:

            src_path = f"{folder_name}/{audio_filename}"
            target_path = f"{dirname}/{audio_filename}"
            #create that folder if it doesn't exist
            if not os.path.isdir(os.path.dirname(target_path)):
                os.mkdir(os.path.dirname(target_path))
            features = extract_feature(src_path, mel=True)
            target_filename = target_path.split(".")[0]
            np.save(target_filename, features)
```

Listing 2: Code to preprocess the dataset

After running this block of code, a new folder named "newdata" should be generated. it should contain 4 .csv files and 3 folders and all the folders should have the .npy files in it. The .csv file named "balanced-all.csv" is the main file which should be used to train the model. This file contains the path to the .npy files and the corresponding label "0" for male or "1" for female. The .npy files contain the mel-spectrograms which are to be used as features to train the model.

3.2.2 Assignments to solve and report

Write the codes for the following tasks in a .ipynb file, including all visualizations and submit the executed file. Submit also a separate pdf of your report on this experiment. This report should describe your observations and reasoning while executing these experiments. Use Tensorflow or PyTorch according to your comfort.

1. Rerun all the steps discussed till now, rerunning all the codes. Make sure that the mentioned folders and files are generated.
2. Create the model architecture.
3. Split the dataset into train, validation and test sets such that 80% of data is used for training and 10% for testing and validation each.
4. Show the size of every tensor in the forward pass.
5. Train it for 100 epochs with a learning rate of 0.001
6. Plot the training loss and accuracy curves.
7. Evaluate the model and show the testing accuracy.
8. Generate a confusion matrix after testing the model on the test set.